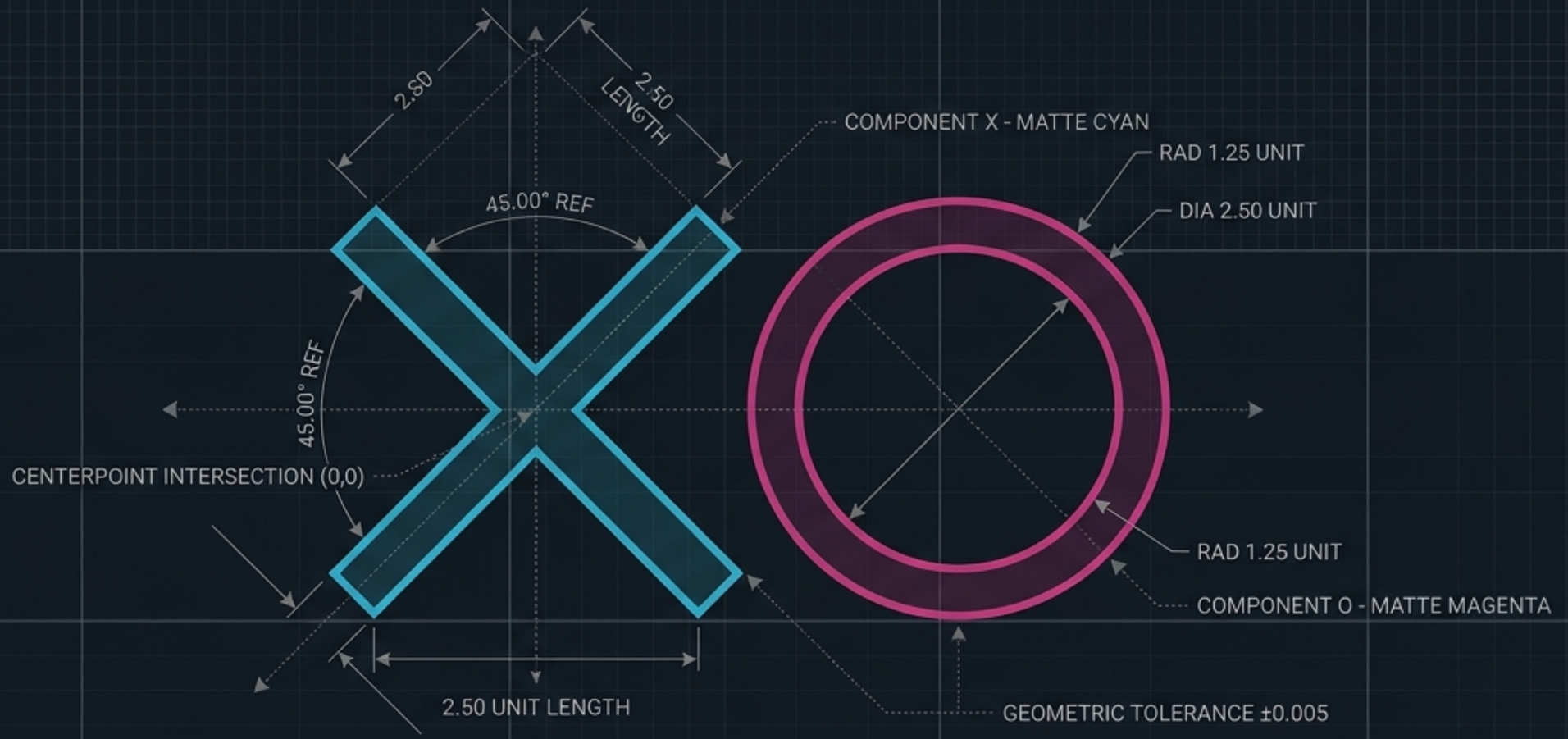


The Architecture of Noughts & Crosses

Domain Modeling, Game Theory, and the MVC Paradigm in a 3x3 Grid

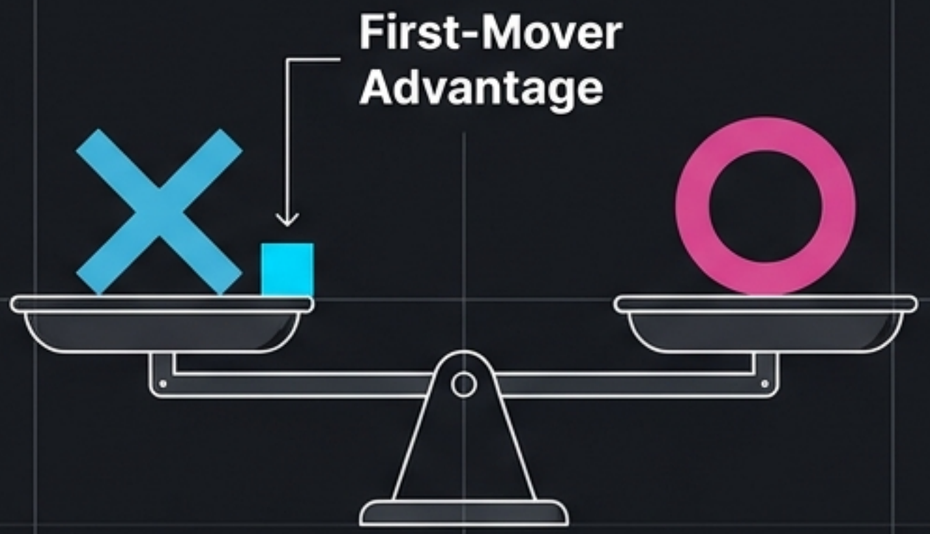
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Adversarial, Zero-Sum, Quasi-Symmetric

1

Adversarial & Zero-Sum

Both players cannot win simultaneously. Play strictly opposes the opponent's victory state.



2

Quasi-Symmetry

Both players rely on the exact same strategy for winning, differing only in their assigned symbol.

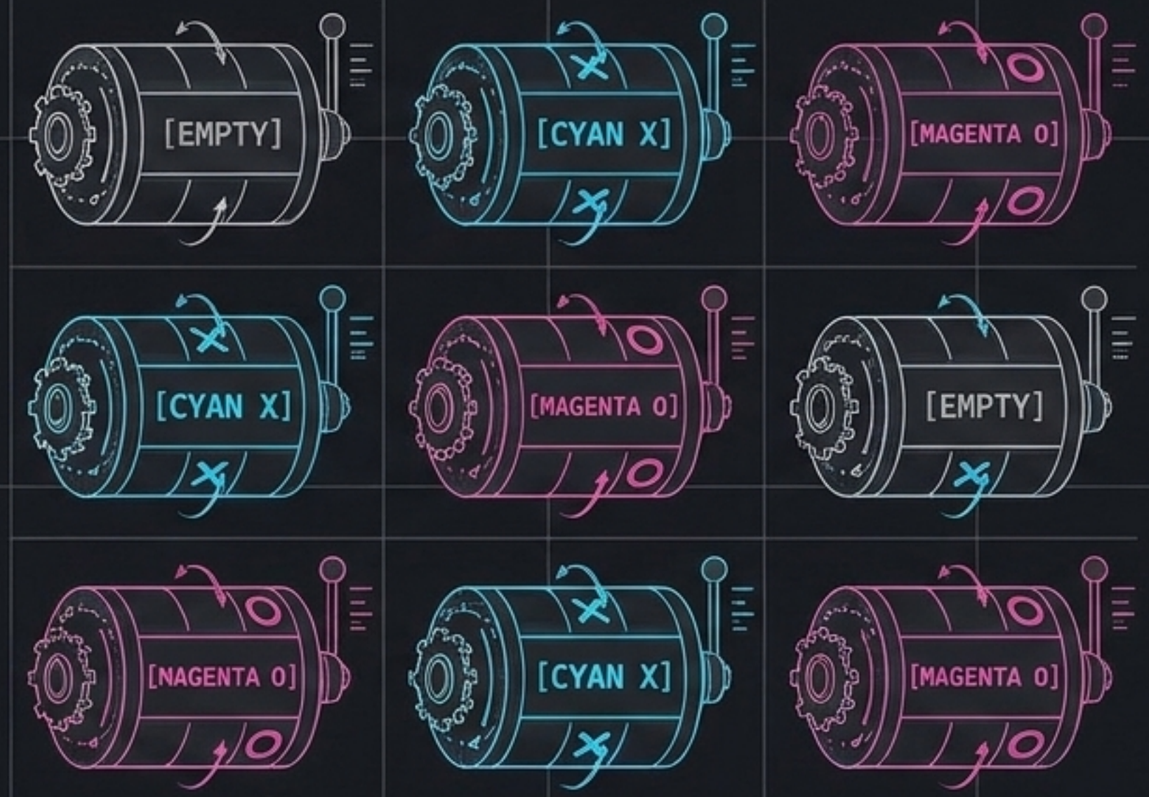
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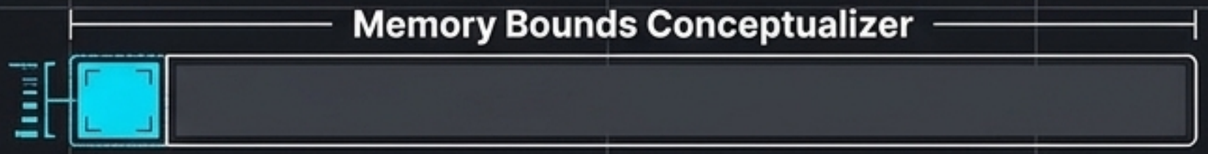
Architectural Implication

Avoid redundant code. Design a single Player class. Instantiate one for P1 and another for P2.

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The Memory Matrix





With only 3 mutually exclusive states per cell, each cell requires just 2 bits of memory.

2 bits × 9 cells = 18 bits total state memory.

1

The Canvas

A 2D array consisting of exactly 9 discrete cells.

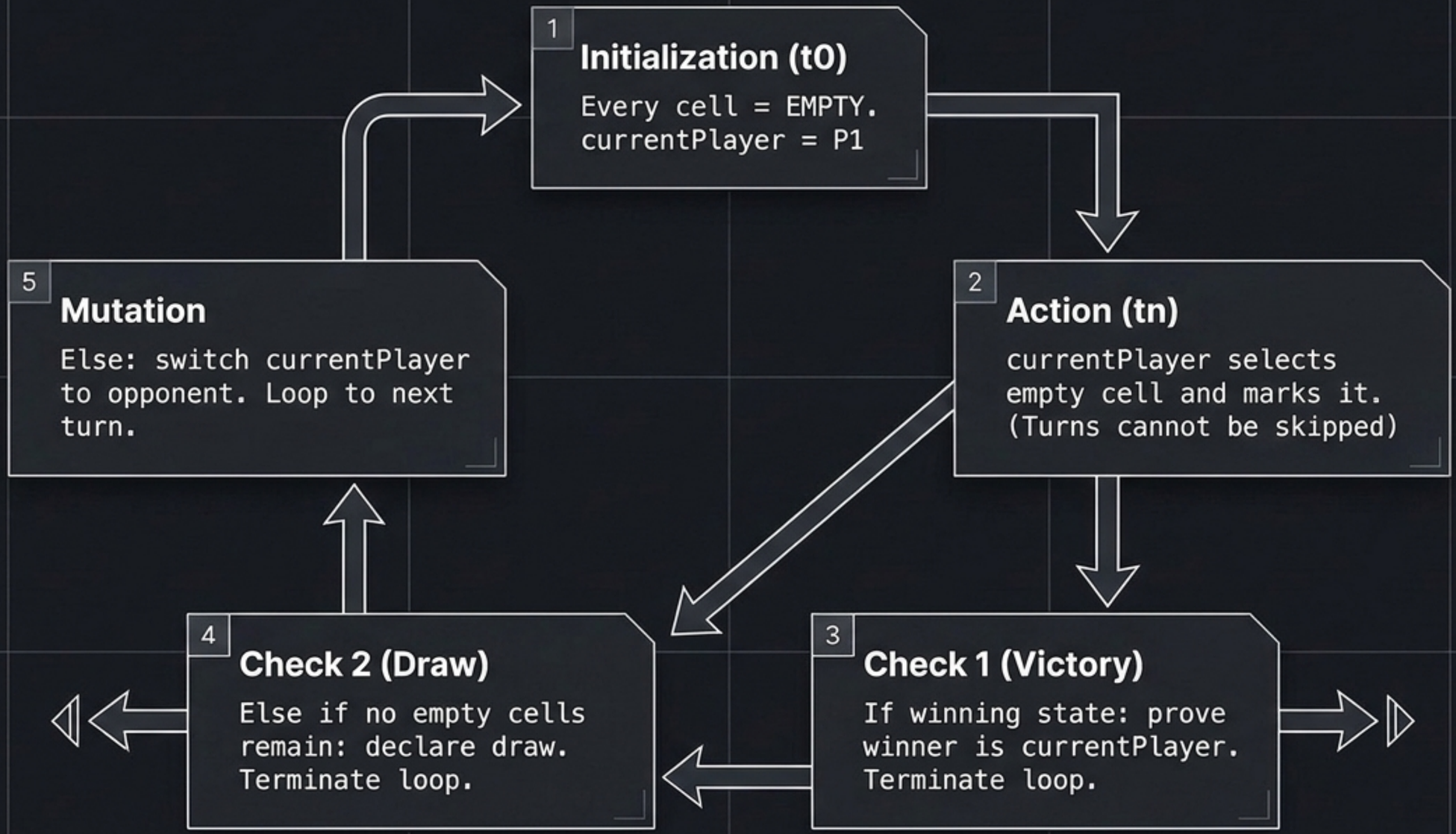
2

Mutually Exclusive States

Every cell exists in exactly one state at any moment: possessing nothing, possessing X, or possessing O.

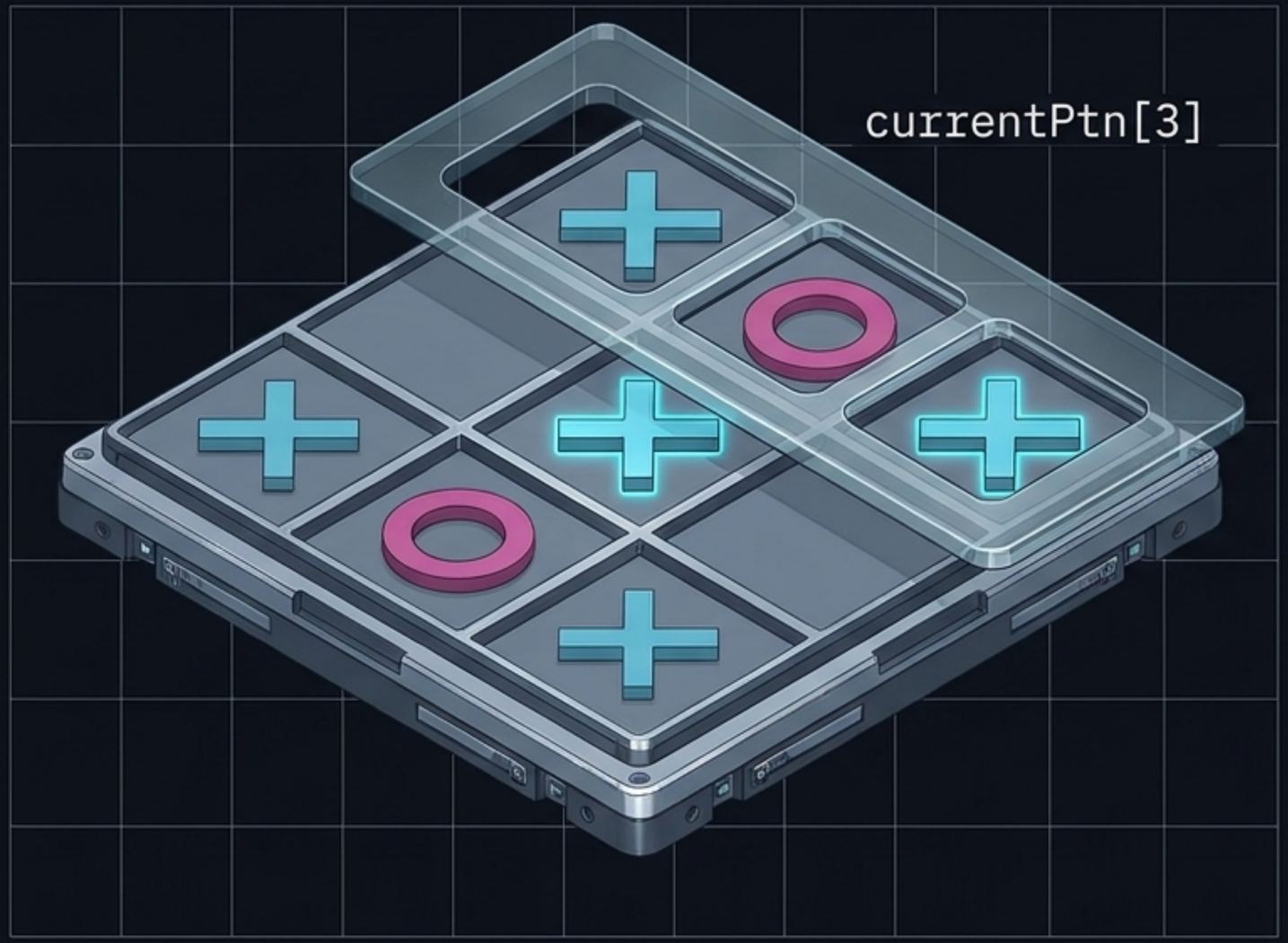
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The Turn-Based State Machine



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Win-Condition Pattern Matching



The Mechanism

Subsystem observes the board at turn-end before updating the turn counter.

Memory Allocation

Allocate a 3-element array: currentPtn[3]

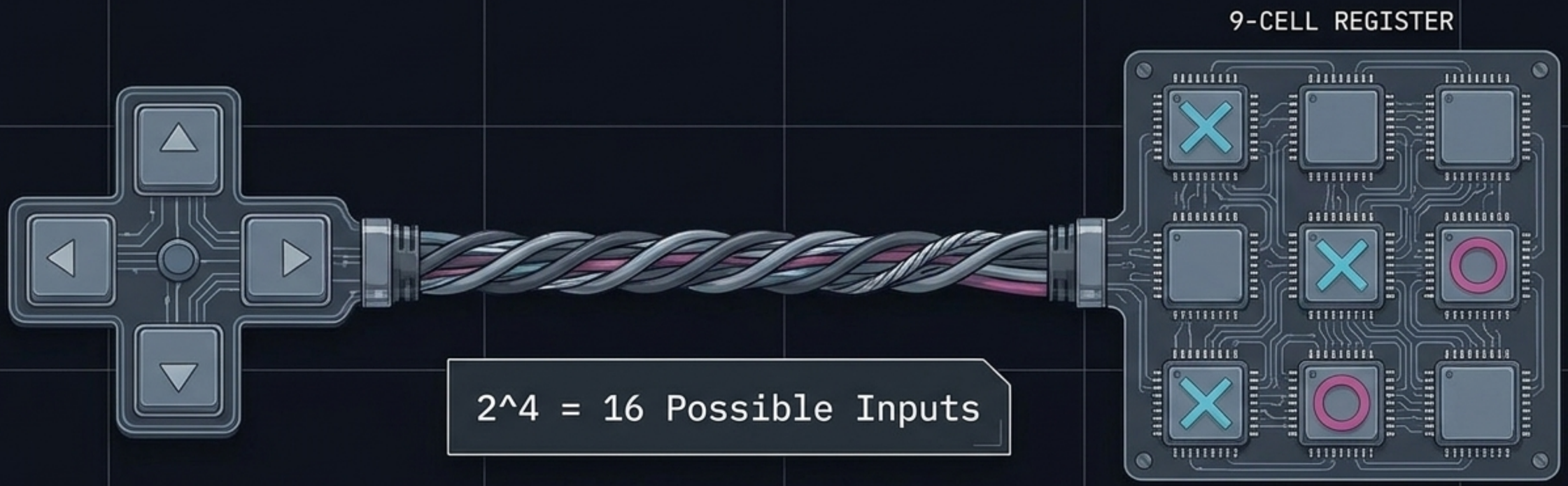
The Logical Proof

To guarantee a win, evaluate every row, column, and diagonal against this rigid Boolean constraint:

```
(currentPtn[0] != EMPTY) AND
(currentPtn[0] == currentPtn[1]) AND
(currentPtn[0] == currentPtn[2])
```

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The 4-Bit Controller Abstraction



The Constraint

The only action available is selecting an empty cell. Marks are statically assigned.

Marks are statically assigned at initialization. Selecting an empty cell inherently means choosing to place a mark there.

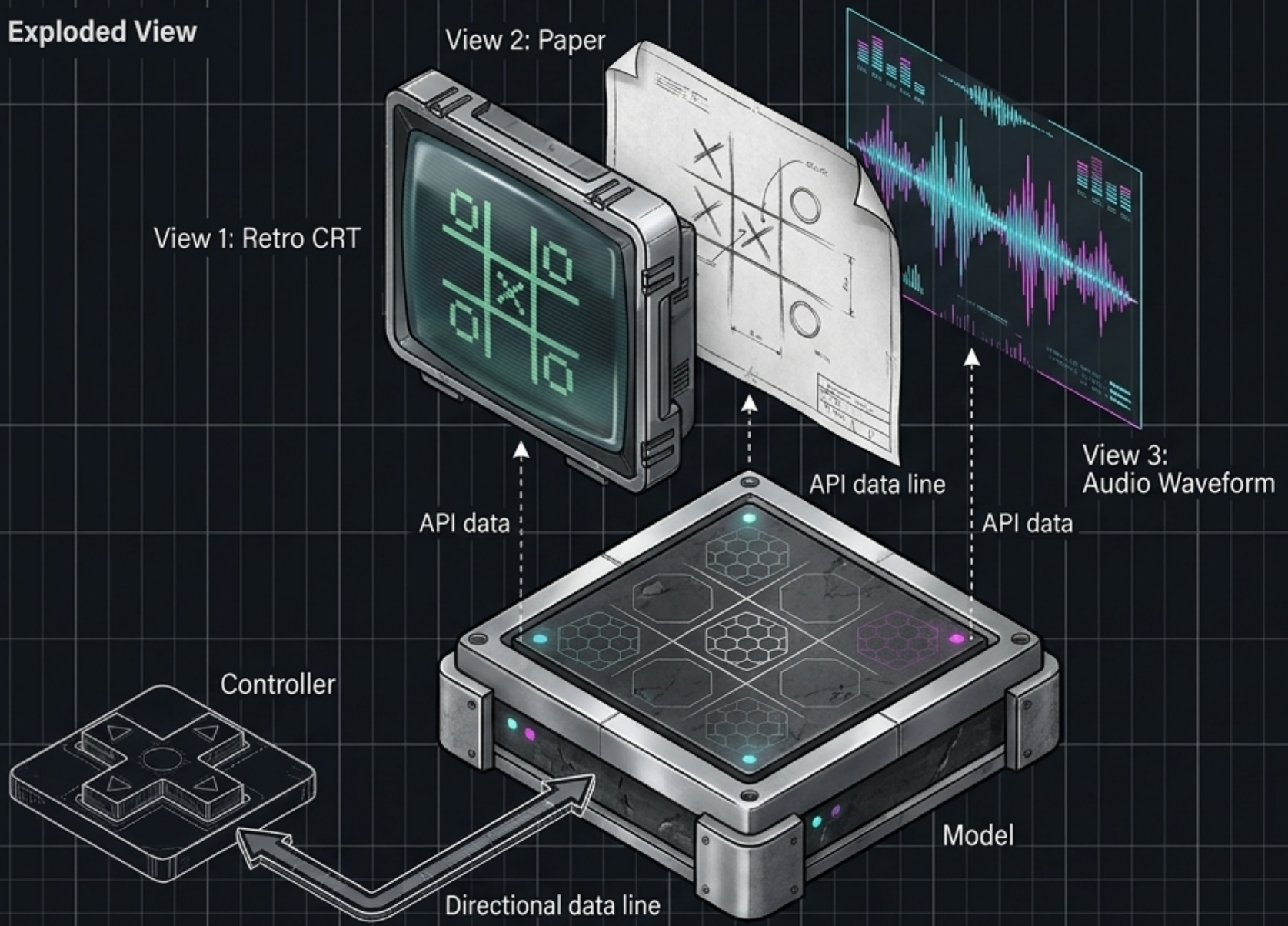
The Implication

Maximum 9 cells.

A 4-bit integer provides 16 unique IDs, covering the board with 7 integers unused. The core system only needs 4 bits to fully represent any player input.

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Decoupling Model from Presentation



The logic of the game is mathematically sealed.

Input representation is hard-capped at 4 bits.

By strictly enforcing the MVC paradigm, the presentation layer is completely untethered from the game logic.

The board is an abstract state; the grid is just an illusion we project onto it.

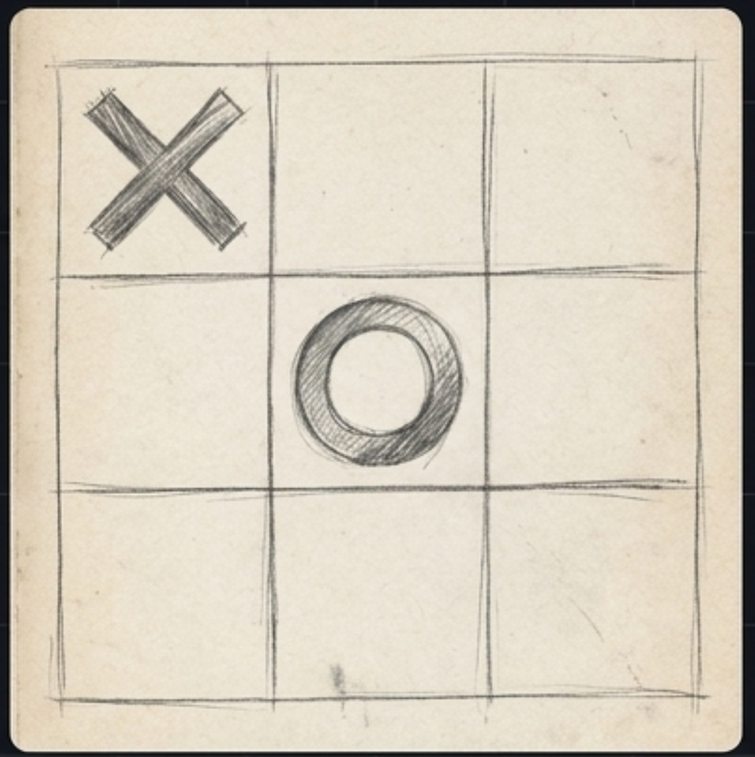
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Visualizing the State



Ghosting UI

Ghosting UI: Rendering a hover-state preview before the 4-bit click event solidifies the mark.



Skeuomorphism

Skeuomorphism: Mapping the abstract math matrix to tangible, real-world materials.



Design Rule

Design Rule: Make X and O distinct; prioritize aesthetic readability over the underlying array structure.

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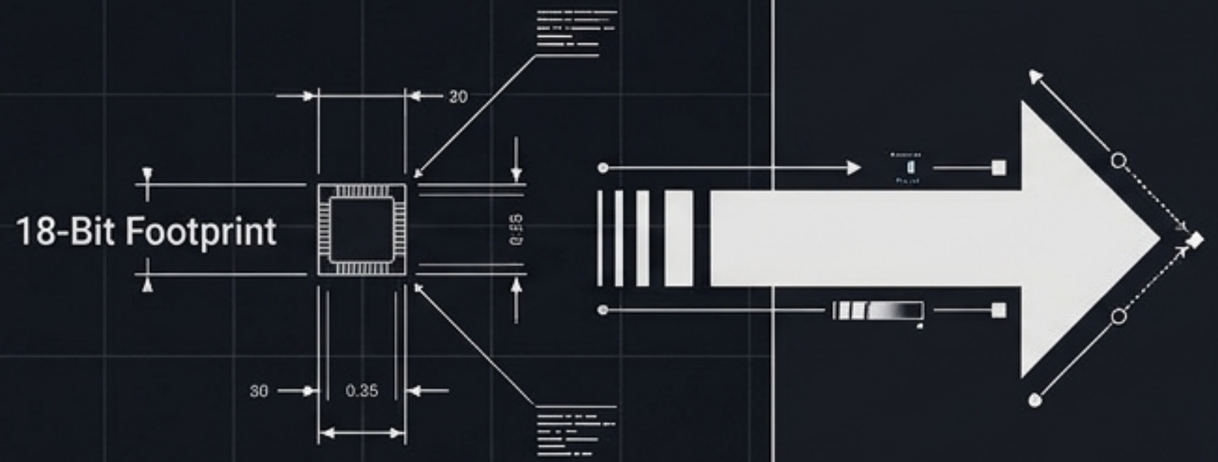
Radical Presentation Modes

Decoupled View Matrix			
Interface Type	**Cognitive Load**	**Implementation Complexity**	**Output Example**
Minimalist (ASCII Art)	Low	Extremely Low	`[X] [ ] [O]`
Graphical (GUI with Ghosting)	Medium	High	
Audio (Speech Synthesis)	High	Medium	***AND THE NOUGHTMASTER PUNCHES A HOLE IN THE BOARD AT CELL (0,2)! OH, THE AGONY! HOW WILL RICK CROSS HANDLE SUCH A DEVASTATING MOVE?!!!***

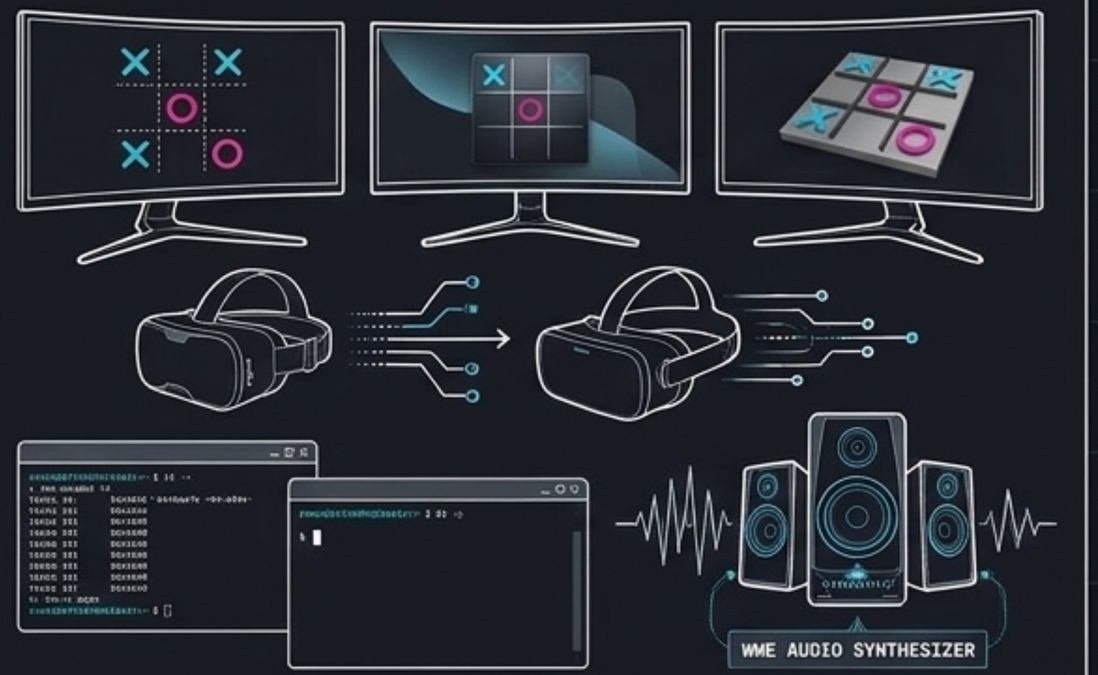
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Rigid Core, Infinite Canvas

The Micro



The Macro



The Thesis:

Strict constraints at the Model layer enable absolute freedom at the Presentation layer.

Because the entire state is

isolated to an 18-bit footprint and mutated via a 4-bit controller, the core logic never changes.

You can output the exact same

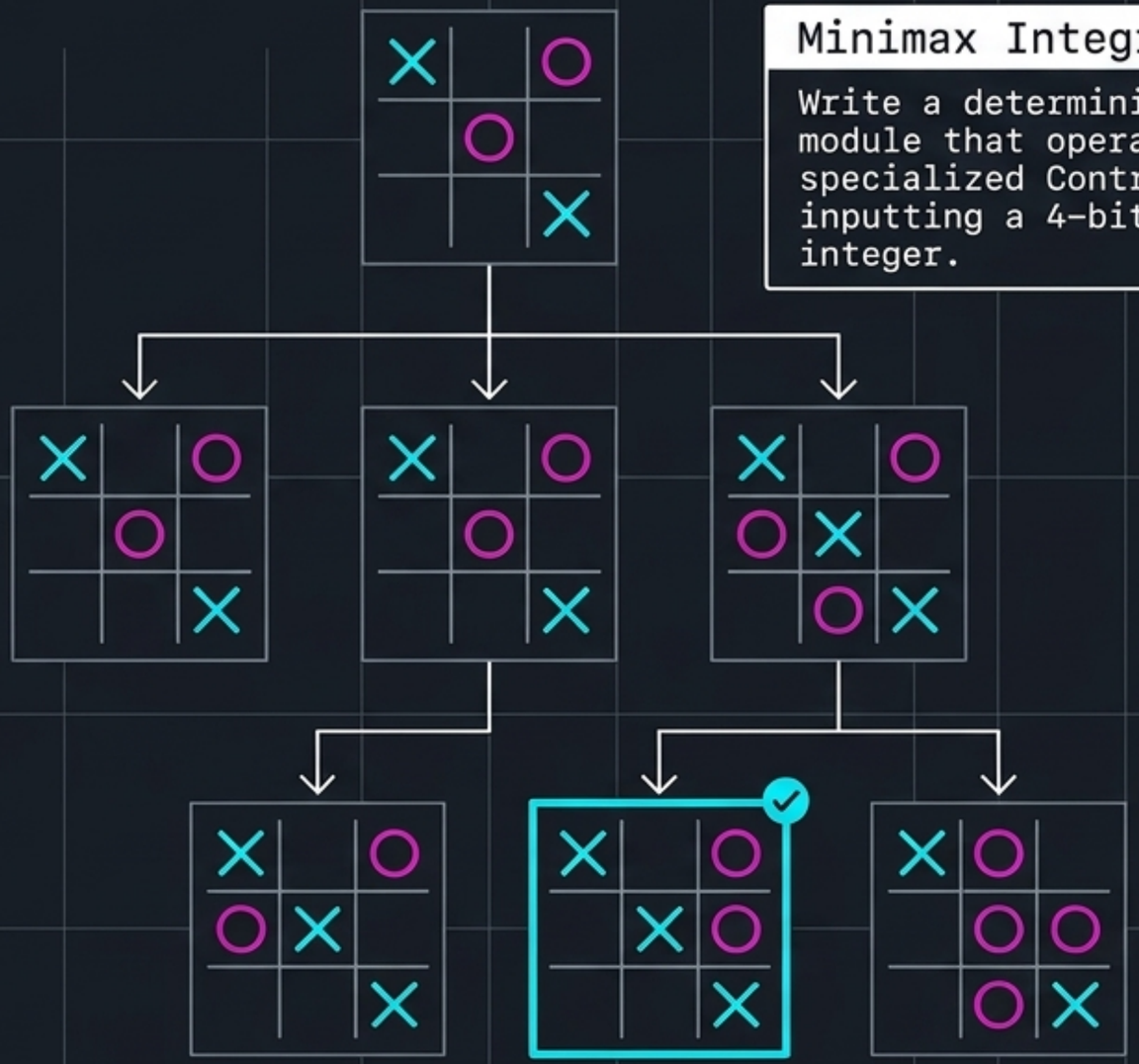
binary state to a terminal, a high-end CGI engine, or a WWE audio synthesizer without modifying a single line of game logic.

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Externalizing Strategy: Deterministic AI

The Zelda Principle

Legend of Zelda was not built in a day. Scale up from flawless small games.



Minimax Integration

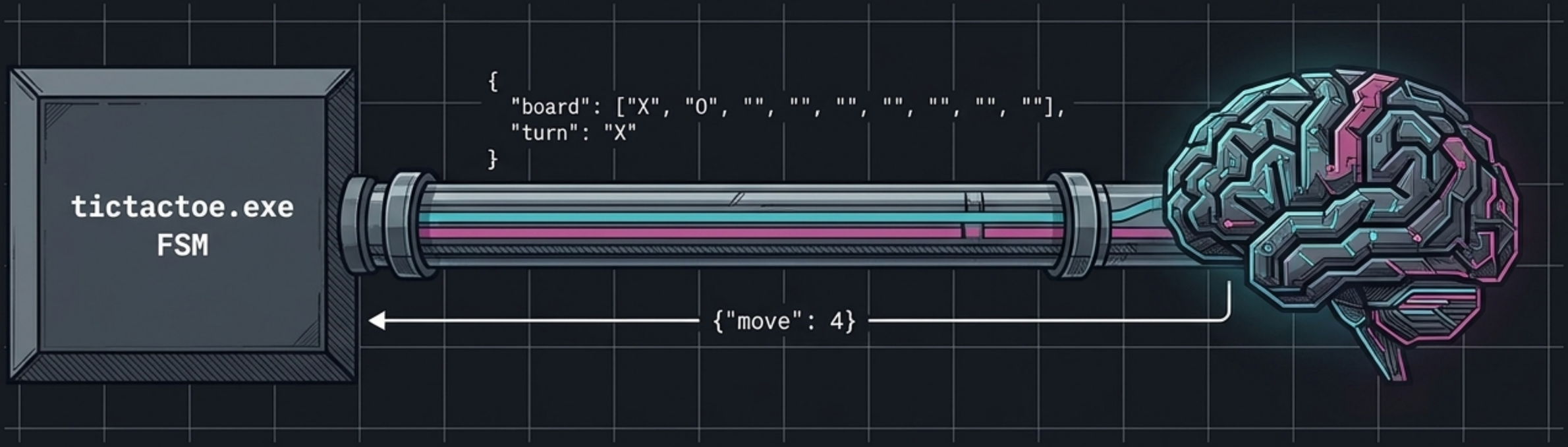
Write a deterministic AI module that operates as a specialized Controller inputting a 4-bit integer.

Predictive Horizon

The AI traverses the State Machine to predict the game two steps in advance, mapping optimal responses to human input.

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Semantic Play: LLM API Integrations



The Interface

Wrapping the rigid FSM in a standardized API layer to accept and broadcast external requests.

The Adversary

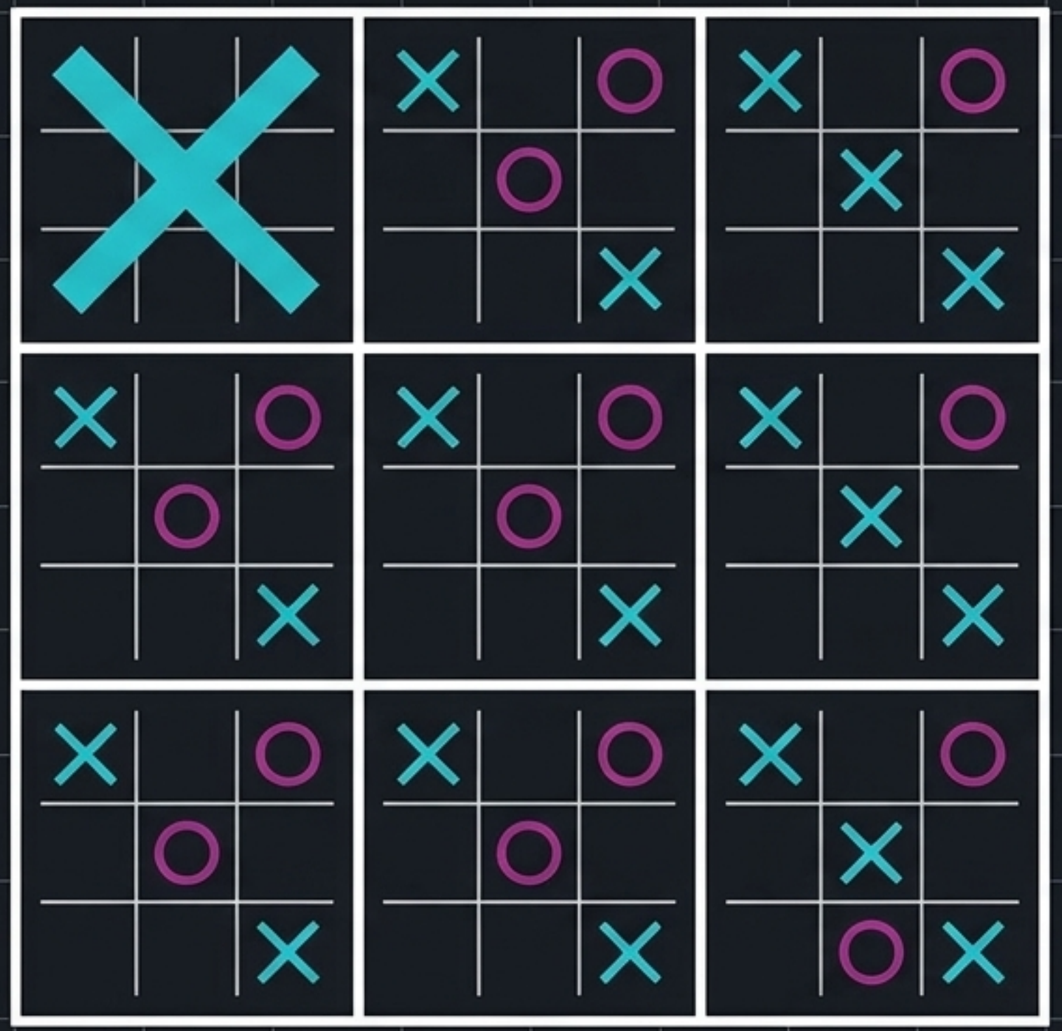
Playing against an LLM shifts the engineering challenge from game logic to prompt engineering and parsing unstructured text.

Multiplayer Scaling

This identical API architecture forms the exact foundation required for network management and persistent tournament systems.

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Fractal Architecture: Mega-Tic-Tac-Toe



Recursive Instantiation

The core 3x3 memory model is instantiated 9 times to construct a macro-board.

Macro Ruleset

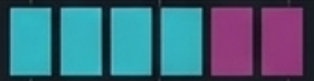
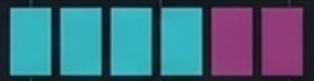
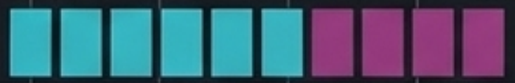
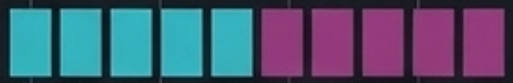
1. Each cell operates as an independent sub-game of TTT.
2. The winner of a sub-game marks the macro-cell with their symbol.
3. Draws result in the macro-cell reverting to the EMPTY state.

Architectural Result

Exponential increase in strategic depth and state complexity using the exact same underlying logic blocks.

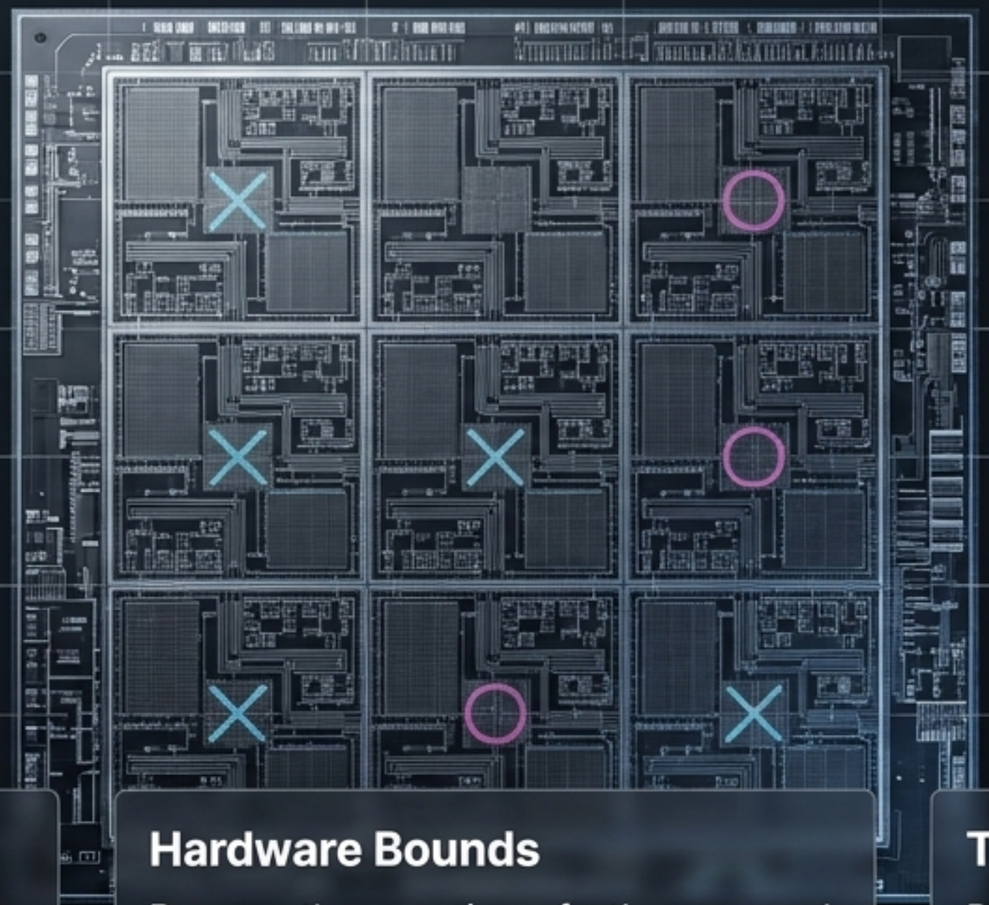
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Evaluating Architectural Extensions Extensibility Matrix

Approach	Predictability	Compute Cost	Dev Effort
Deterministic AI (Minimax)	 Absolute	 Low	 High (Math Heavy)
LLM API Integration	 Variable	 High	 Medium (Parsing Heavy)
Ultimate Mega-TTT	 Perfect	 Low	 High (State Heavy)

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Master the Micro, Architect the Macro



N-by-N Generalization

Discovering optimal play strategies and efficient pattern-matching for infinitely scaling board dimensions.

Hardware Bounds

Because the game is perfectly expressed as an FSM, maximum memory usage is absolutely predictable, enabling tiny self-contained bootloader implementations.

The Takeaway

Rigorous domain modeling and enforced decoupling transforms seemingly trivial problems into mathematically unbreakable systems.

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